

Thenmozhi Boopathy

thenmozhi@boopathy.dev | +1513 331 1208 | github.com/boopatt/ | www.linkedin.com/in/boopatt/ | thenmozhi.boopathy.dev

PROFESSIONAL SUMMARY

MS Business Analytics candidate at Miami University and former mathematics professor, bringing hands-on experience building data pipelines, ML workflows, and cloud-based analytics solutions on Databricks. Applied end-to-end workflows spanning classification, forecasting, deep learning, and agentic RAG using Python, R, and Apache Spark. Databricks University Alliance certified with advanced degrees in mathematics.

EDUCATION

Miami University <i>Master of Science in Business Analytics</i>	Oxford, OH Expected Graduation 2027
Bharathidasan University <i>Master of Philosophy in Mathematics</i>	TamilNadu, India 2008
<i>Master of Science in Mathematics</i>	2007
<i>Bachelor of Science in Mathematics</i>	2005

RELEVANT COURSEWORK

Data Mining: Exploratory Data Analysis, Dimension Reduction, Tree-Based Models, Logistic Regression, Neural Networks, Model Evaluation,
Big Data: Core Python, Pandas, NoSQL (MongoDB), Web APIs, Scikit-learn, Spark MLlib, Text Mining (NLTK, TF-IDF, Word2Vec).
Cloud & GenAI: Databricks, Apache Spark, Delta Lake, Unity Catalog, Databricks Vector Search, LangChain, MLflow, Agentic AI, RAG.

PROJECT EXPERIENCE

Enterprise RAG Chatbot — Job Finder Assistant (ISA-632)	2026
Built and deployed an enterprise-grade Retrieval-Augmented Generation (RAG) chatbot on Databricks with a 3-person team: used Apache Spark to ingest 2,500+ job records into Delta Tables, integrated Databricks Vector Search with a LangChain agent on Llama 4 Maverick, and built a Streamlit UI with MLflow-tracked evaluation, achieving 89% positive user feedback.	
End-to-End R Machine Learning Workflow for Loan Default Prediction (ISA-591)	2025
Built an end-to-end R pipeline (decision trees, logistic regression, neural network, XGBoost) to predict loan default on an imbalanced dataset (80.8% no-default / 19.2% default); diagnosed class imbalance behind a 0.069 baseline sensitivity, then raised it to 0.64 (AUC 0.74) via downsampling and cross-validated pruning, and tuned the decision threshold with Youden's J to minimize costly false negatives.	
Theme Park Attendance Forecasting Pipeline with Python (ISA-514)	2025
Built a Python-based machine learning pipeline using MongoDB for data storage and integration of structured and unstructured sources (weather, wait times, schedules, Reddit sentiment) to forecast Six Flags attendance for operational decision-making.	

PROFESSIONAL EXPERIENCE

S.A College of Engineering <i>Assistant Professor</i>	Chennai, India Jan–May 2015
Delivered instruction in quantitative and statistical methods (numerical analysis, probability, matrices, and applied functions), emphasizing analytical problem-solving skills relevant to data-driven business decision-making.	
Sri Venkateswara College of Engineering <i>Assistant Professor</i>	Chennai, India 2012-2013
Performed lecturing in advanced quantitative methods (calculus, sequences, integrals), building strong analytical and problem-solving foundations relevant to data modeling and forecasting.	
Bon Secours College for Women <i>Lecturer (PG and UG)</i>	Tanjore, India 2009-2012
Conducted training in mathematics and analytics concepts for undergraduate and postgraduate students, strengthening quantitative reasoning, data interpretation, and problem-solving skills.	

VOLUNTEER ACTIVITIES

Language Teacher <i>Greater Cincinnati Tamil School, Cincinnati</i>	Cincinnati, Ohio 2021-2025
Designed and provided customized language programs for students of varying proficiency levels, using assessments to track progress and enhance outcomes. Coordinated program logistics and engaged stakeholders to promote the Seal of Bilingualism.	
Gifted Education Co-Representative <i>Sycamore Gear</i>	Cincinnati, Ohio 2023-2024
Designed and executed student engagement initiatives, enhancing community involvement and program effectiveness.	

SKILLS

Certification: Databricks University Alliance Student — Databricks (issued 2026). TNSET (Tamil Nadu State Eligibility Test) — Assistant Professor Eligibility, Mathematical Sciences (2017).

Languages & Tools: Python, R, SQL, Jupyter Notebook, RStudio, Visual Studio Code, GIT

Data Platforms: Databricks, Apache Spark, Spark MLlib, Delta Lake, NoSQL (MongoDB), Power BI

ML / AI: Pandas, Data & ML Pipelines, LangChain, MLflow, Vector Search, GenAI / LLM, Agentic AI, Deep Learning, NLTK